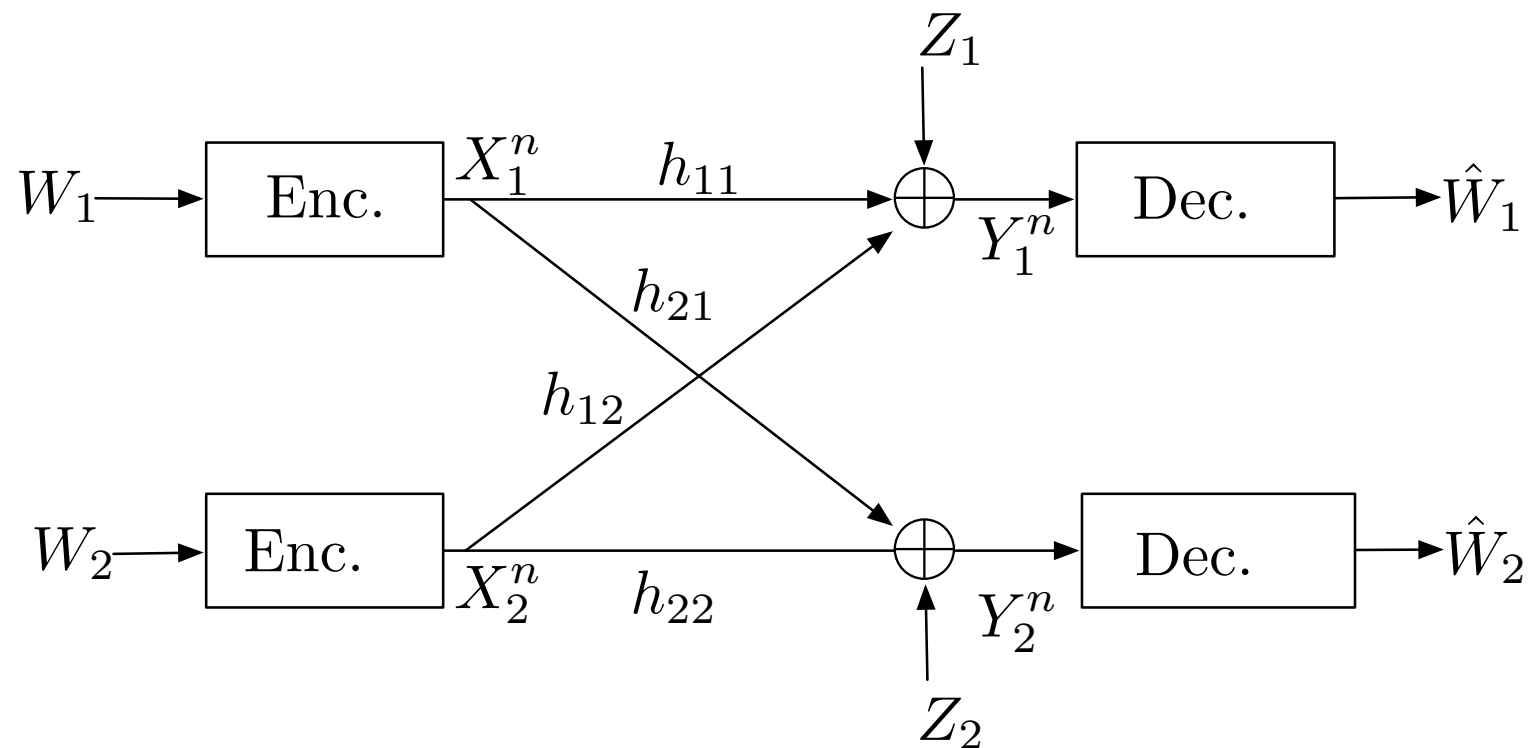


Nearly Optimal Non-Gaussian Codes for the Gaussian Interference Channel

Alex Dytso, Daniela Tuninetti, Natasha Devroye

Channel Model



with the usual assumptions and
capacity region definition.

Symmetric:
$$\begin{aligned} |h_{11}|^2 &= |h_{22}|^2 = S \\ |h_{12}|^2 &= |h_{21}|^2 = I \end{aligned}$$

Motivation

Capacity: $\mathcal{C} = \lim_{n \rightarrow \infty} \text{co} \left(\bigcup_{P_{X_1^n X_2^n} = P_{X_1^n} P_{X_2^n}} \left\{ \begin{array}{l} 0 \leq R_1 \leq \frac{1}{n} I(X_1^n; Y_1^n) \\ 0 \leq R_2 \leq \frac{1}{n} I(X_2^n; Y_2^n) \end{array} \right\} \right)$

R. Ahlswede, "Multi-way communication channels," in Proc. IEEE Int. Symp. Inf. Theory, March 1973, pp. 23–52.

Treat Interference as Noise Inner Bound:

$$\mathcal{R}_{\text{in}}^{\text{TIN+TS}} = \text{co} \left(\bigcup_{P_{X_1 X_2} = P_{X_1} P_{X_2}} \left\{ \begin{array}{l} 0 \leq R_1 \leq I(X_1; Y_1) \\ 0 \leq R_2 \leq I(X_2; Y_2) \end{array} \right\} \right) \quad \text{With Time Sharing}$$

$$\mathcal{R}_{\text{in}}^{\text{TINnoTS}} = \bigcup_{P_{X_1 X_2} = P_{X_1} P_{X_2}} \left\{ \begin{array}{l} 0 \leq R_1 \leq I(X_1; Y_1) \\ 0 \leq R_2 \leq I(X_2; Y_2) \end{array} \right\} \quad \text{No Time Sharing}$$

How far away is TINnoTS from the Capacity?

Outline

- Definitions and relevant past work
- Useful tools:
 - Mutual information lower bounds
 - Minimum Distance lower bounds
- Main result

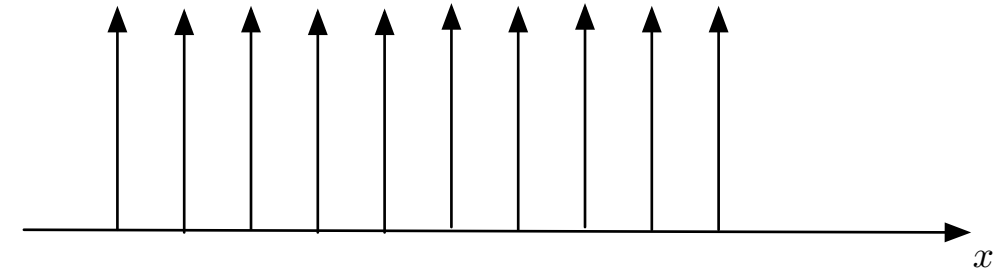
Mixed Inputs

$$X = \sqrt{1 - \delta} X_D + \sqrt{\delta} X_G,$$

$$\delta \in [0, 1],$$

$$X_D \sim \text{PAM}(N),$$

$$X_G \sim \mathcal{N}(0, 1)$$



PAM

$$X = \sqrt{1 - \delta} X_{D_1} + \sqrt{\delta} X_{D_2}$$

$$X_{D_1} \sim \text{PAM}(N_1), \quad X_{D_2} \sim \text{PAM}(N_2)$$

Known Results

Capacity Strong Interference

$$|h_{11}|^2 \leq |h_{21}|^2 \text{ and } |h_{22}|^2 \leq |h_{12}|^2$$

Sato, H., "The capacity of the Gaussian interference channel under strong interference," IEEE Trans. Inf. Theory, vol. 27, no. 6, pp. 786,788, Nov 1981.

Sum-Capacity in Very Weak Interference

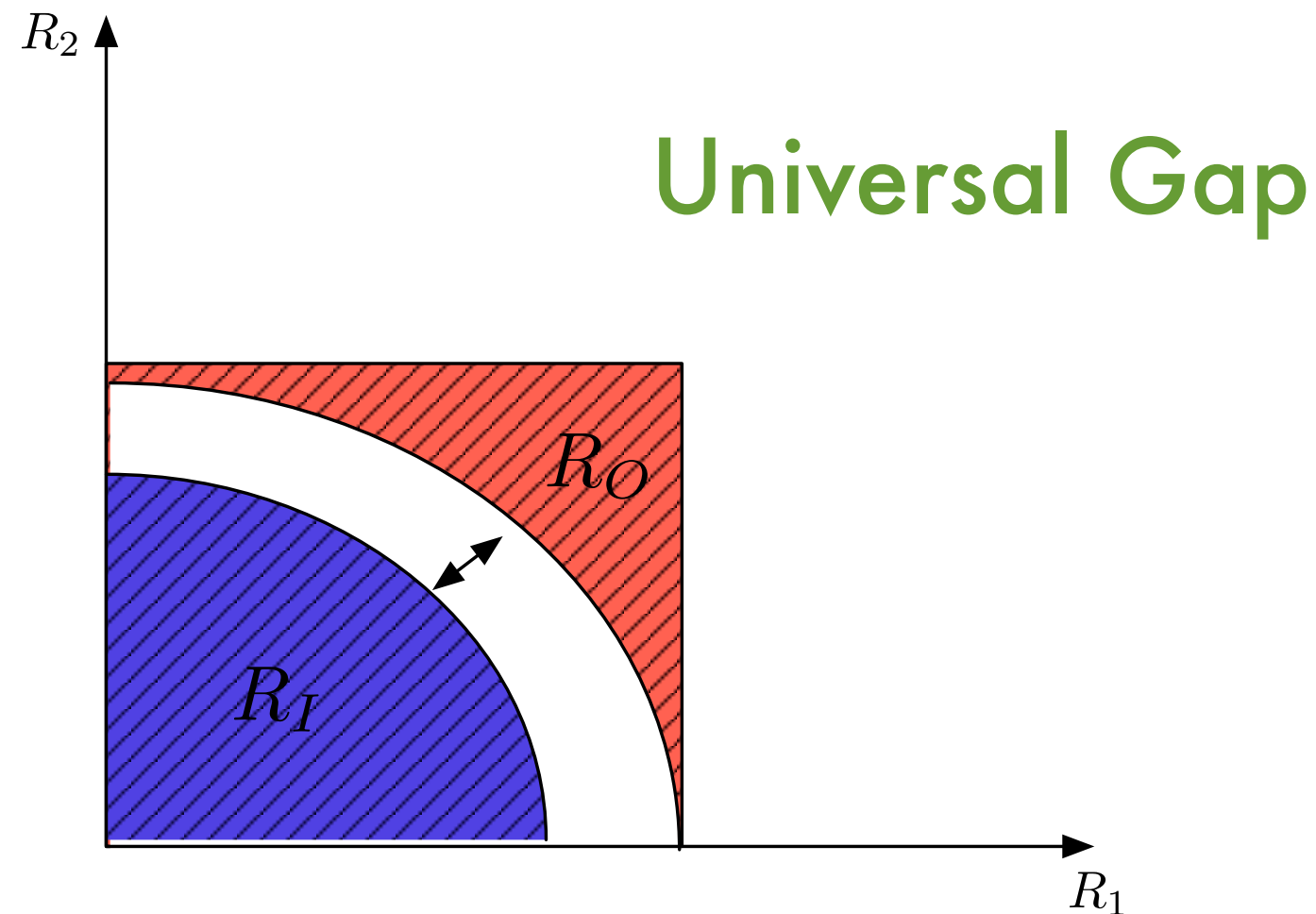
$$\sqrt{\frac{S}{I}} (1 + I) \leq \frac{1}{2}$$

X. Shang, G. Kramer, and B. Chen, "A new outer bound and the noisy-interference sum rate capacity for Gaussian interference channels," IEEE Trans. Inf. Theory, vol. 55, no. 2, pp. 689–699, 2009.

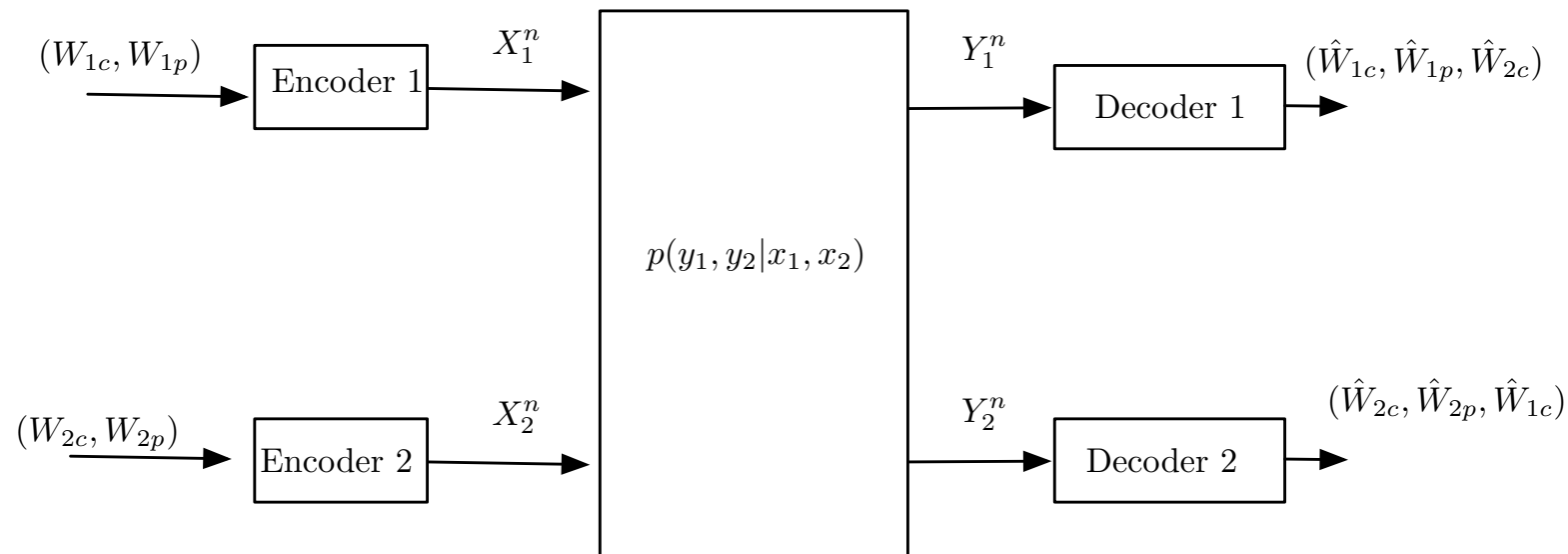
Motahari, A.S.; Khandani, A.K., "Capacity Bounds for the Gaussian Interference Channel," IEEE Trans. Inf. Theory, vol.55, no.2, pp.620,643, Feb. 2009

Annapureddy, V.S.; Veeravalli, V.V., "Gaussian Interference Networks: Sum Capacity in the Low-Interference Regime and New Outer Bounds on the Capacity Region," IEEE Trans. Inf. Theory, vol.55, no. 7, pp.3032,3050, July 2009

Approximate Capacity

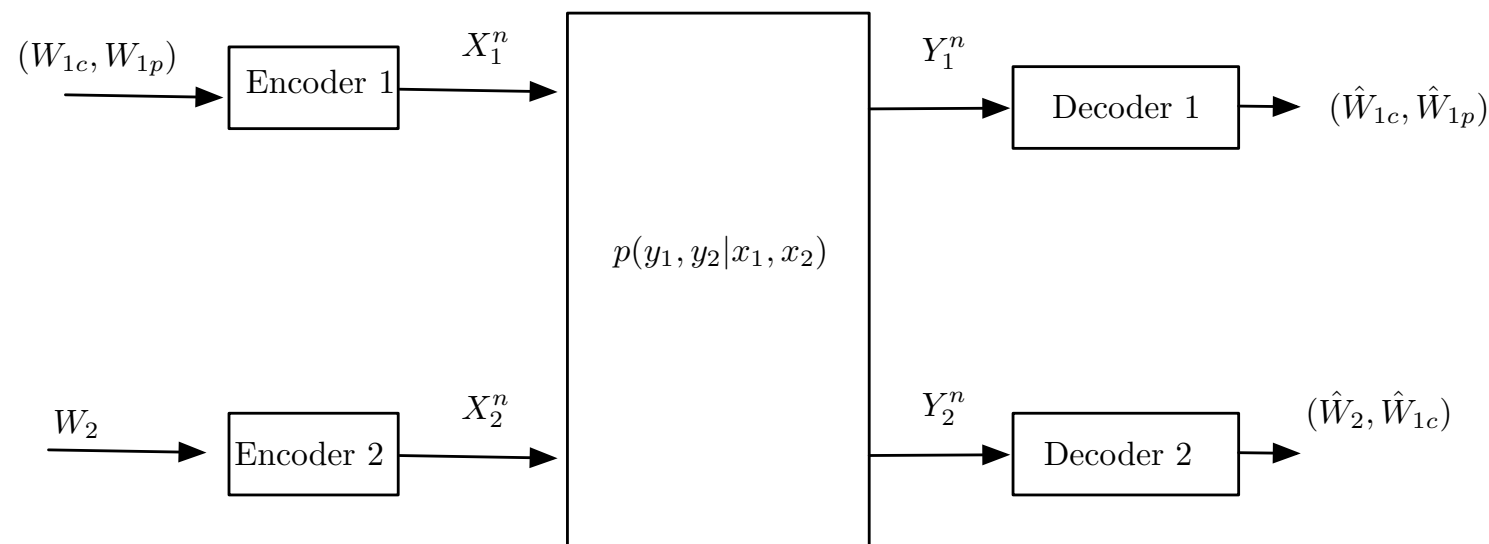


Approximate Capacity



HK+Gaussian Inputs
1/2 bit

R. Etkin, D. Tse, and H. Wang, "Gaussian interference channel capacity to within one bit," IEEE Trans. Inf. Theory, vol. 54, no. 12, pp. 5534–5562, Dec. 2008.

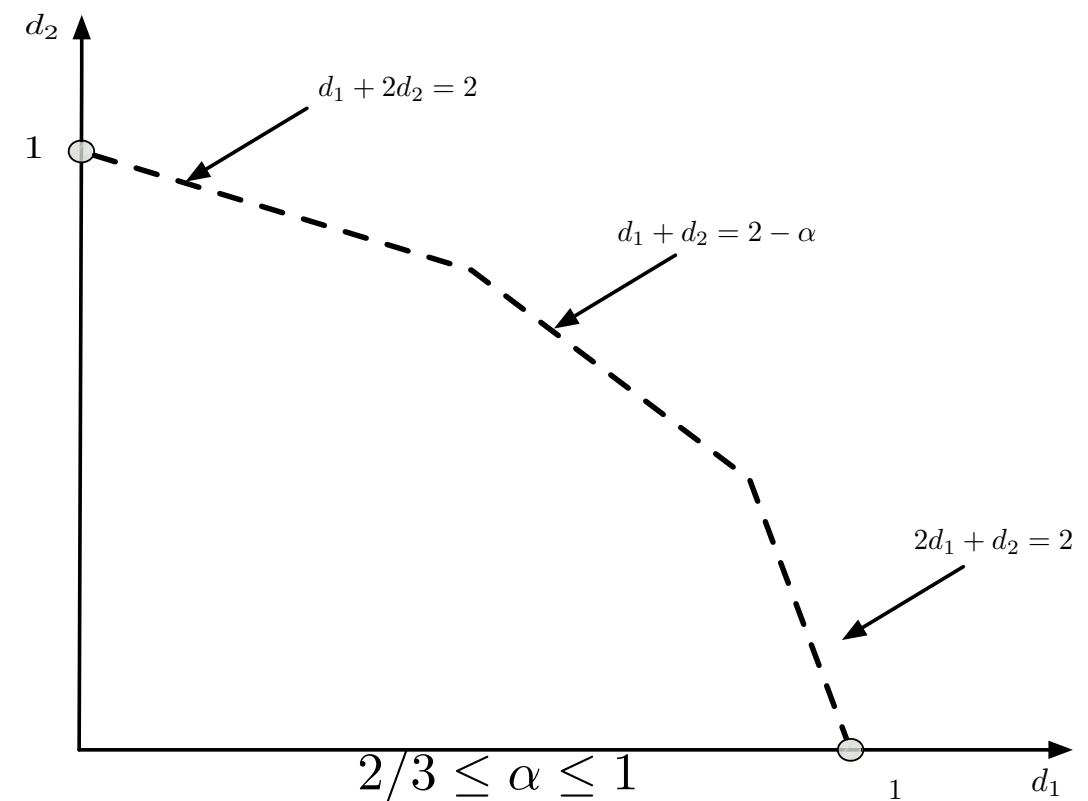
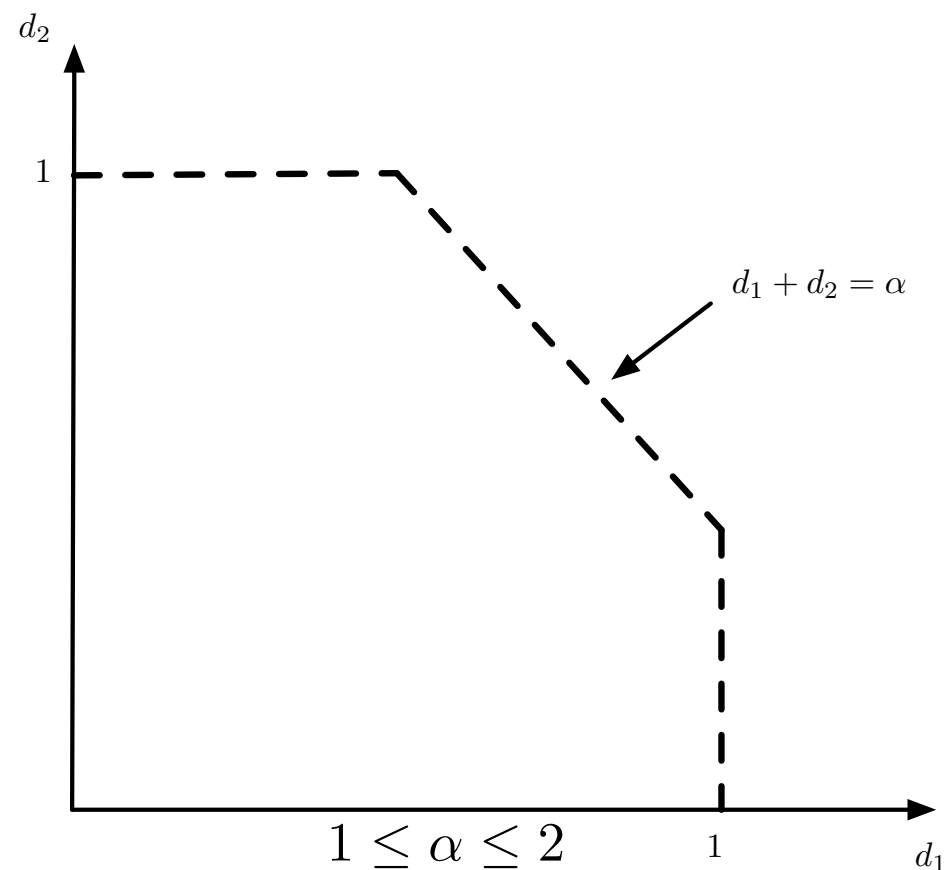


**"One-sided" HK+
Mixed Inputs**
1.75 bits

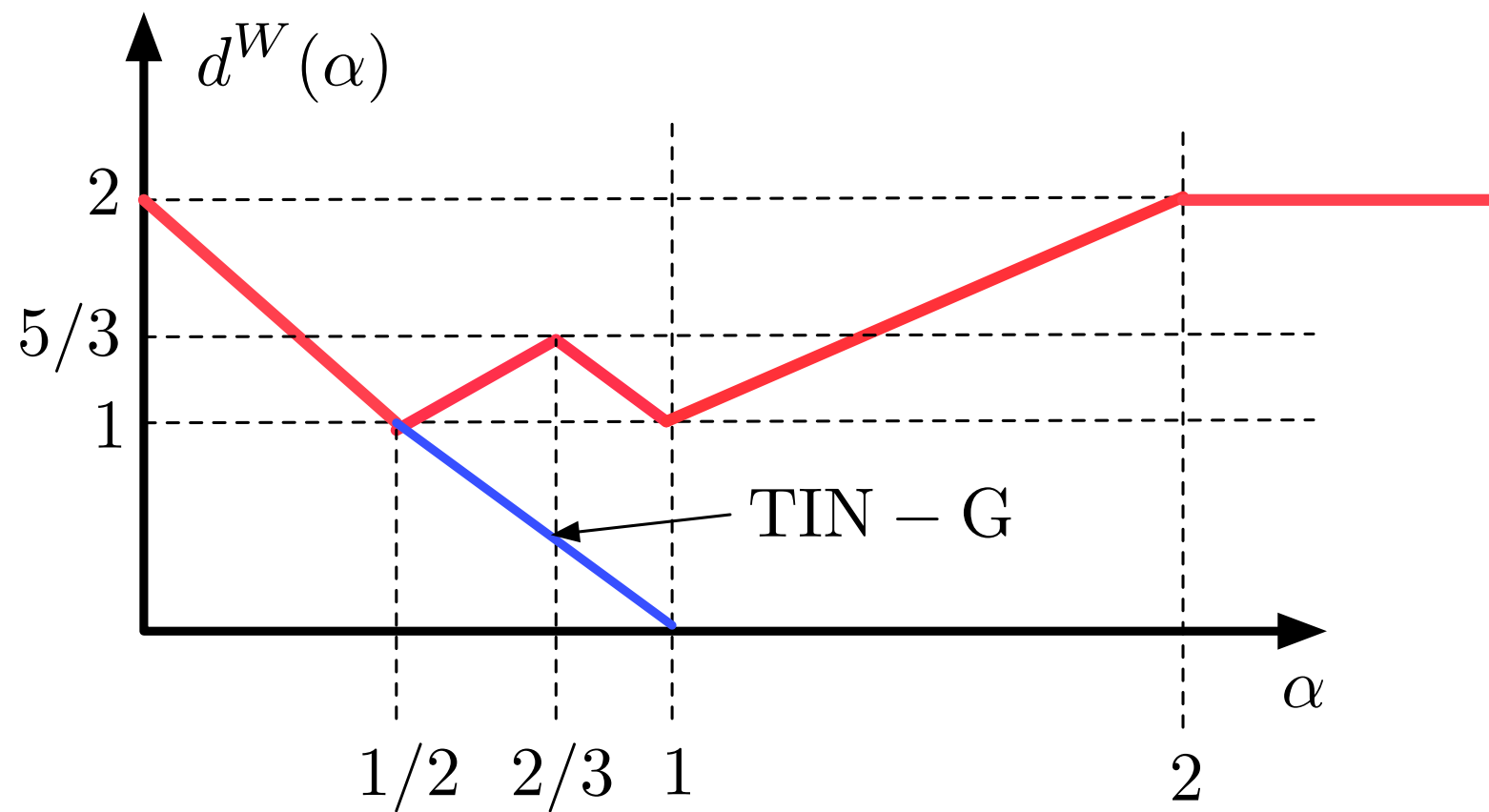
A. Dytso, D. Tuninetti, and N. Devroye, "On the two-user interference channel with lack of knowledge of the interference codebook at one receiver," IEEE Trans. Inf. Theory, vol. 61, no. 3, pp. 1257–1276, March 2015.

gDoF Region

$$\mathcal{D}(\alpha) := \left\{ (d_1, d_2) \in \mathbb{R}_+^2 : d_i := \lim_{\substack{I = S^\alpha, \\ S \rightarrow \infty}} \frac{R_i}{\frac{1}{2} \log(1 + S)}, i \in [1 : 2], (R_1, R_2) \text{ is achievable} \right\}.$$



Sum-gDoF



$$\mathcal{R}_{\text{in}}^{\text{TINnoTS}} = \bigcup_{P_{X_1} P_{X_2}} \left\{ \begin{array}{l} 0 \leq R_1 \leq I(X_1; Y_1) \\ 0 \leq R_2 \leq I(X_2; Y_2) \end{array} \right\}$$

Achievability

$$\mathcal{R}_{\text{in}}^{\text{TINnoTS}} = \bigcup_{N_1, N_2, \delta_1, \delta_2} \left\{ \begin{array}{l} 0 \leq R_1 \leq I(X_1; Y_1) \\ 0 \leq R_2 \leq I(X_2; Y_2) \end{array} \right\} \quad \text{with} \quad \begin{array}{l} X_i = \sqrt{1 - \delta_i} X_{iD} + \sqrt{\delta_i} X_{iG}, \\ \delta_i \in [0, 1], \\ X_{iD} \sim \text{PAM}(N_i), \\ X_{iG} \sim \mathcal{N}(0, 1), \\ i = 1, 2. \end{array}$$

Analytical rate expressions not trivial

$$\begin{aligned} I(X_2; Y_2) &= I(X_2; h_{21}X_1 + h_{22}X_2 + Z_G) \\ &= I\left(X_{1D}, X_{2D}; \frac{\sqrt{1 - \delta_1}h_{21}X_{1D} + \sqrt{1 - \delta_2}h_{22}X_{2D}}{\sqrt{1 + |h_{21}|^2\delta_1 + |h_{22}|^2\delta_2}} + Z_G\right) \\ &\quad - I\left(X_{1D}; \frac{\sqrt{1 - \delta_1}}{\sqrt{1 + |h_{21}|^2\delta_1}}h_{21}X_{1D} + Z_G\right) \\ &\quad + \frac{1}{2} \log(1 + |h_{21}|^2\delta_1 + |h_{22}|^2\delta_2) - \frac{1}{2} \log(1 + |h_{21}|^2\delta_1). \end{aligned}$$

Bounds

Ozarow-Wyner-A

$$I(X_D; \sqrt{S}X_D + Z) \geq H(X_D) - \text{gap},$$

$$\text{gap} = \xi \log \frac{1}{\xi} + (1 - \xi) \log \frac{1}{1 - \xi} + \xi \log(N - 1),$$

$$\xi := 2Q \left(\frac{\sqrt{S}d_{\min}(X_D)}{2} \right),$$

Ozarow-Wyner-B

$$I(X_D; \sqrt{S}X_D + Z) \geq H(X_D) - \text{gap},$$

$$\text{gap} = \frac{1}{2} \log \left(\frac{\pi e}{6} \right) + \frac{1}{2} \log \left(1 + \frac{12}{Sd_{\min}^2(X_D)} \right).$$

DTD-ITA`14-A

$$I(X_D; \sqrt{S}X_D + Z)$$

$$\geq \left[-\log \left(\sum_{(i,j) \in [1:N]^2} p_i p_j \frac{1}{\sqrt{4\pi}} e^{-\frac{(s_i - s_j)^2}{4}} \right) - \frac{1}{2} \log(2\pi e) \right]^+$$

DTD-ITA`14-B

$$I(X_D; \sqrt{S}X_D + Z) \geq \log(N) - \text{gap},$$

$$\text{gap} = \frac{1}{2} \log \left(\frac{e}{2} \right) + \frac{1}{2} \log \left(1 + (N - 1) e^{-\frac{Sd_{\min}^2(X_D)}{4}} \right)$$

Bounds

Ozarow-Wyner-A

$$I(X_D; \sqrt{S}X_D + Z) \geq H(X_D) - \text{gap},$$

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$$\xi := 2Q \left(\frac{\sqrt{S}d_{\min}(X_D)}{2} \right),$$

Ozarow-Wyner-B

$$I(X_D; \sqrt{S}X_D + Z) \geq H(X_D) - \text{gap},$$

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DTD-ITA`14-A

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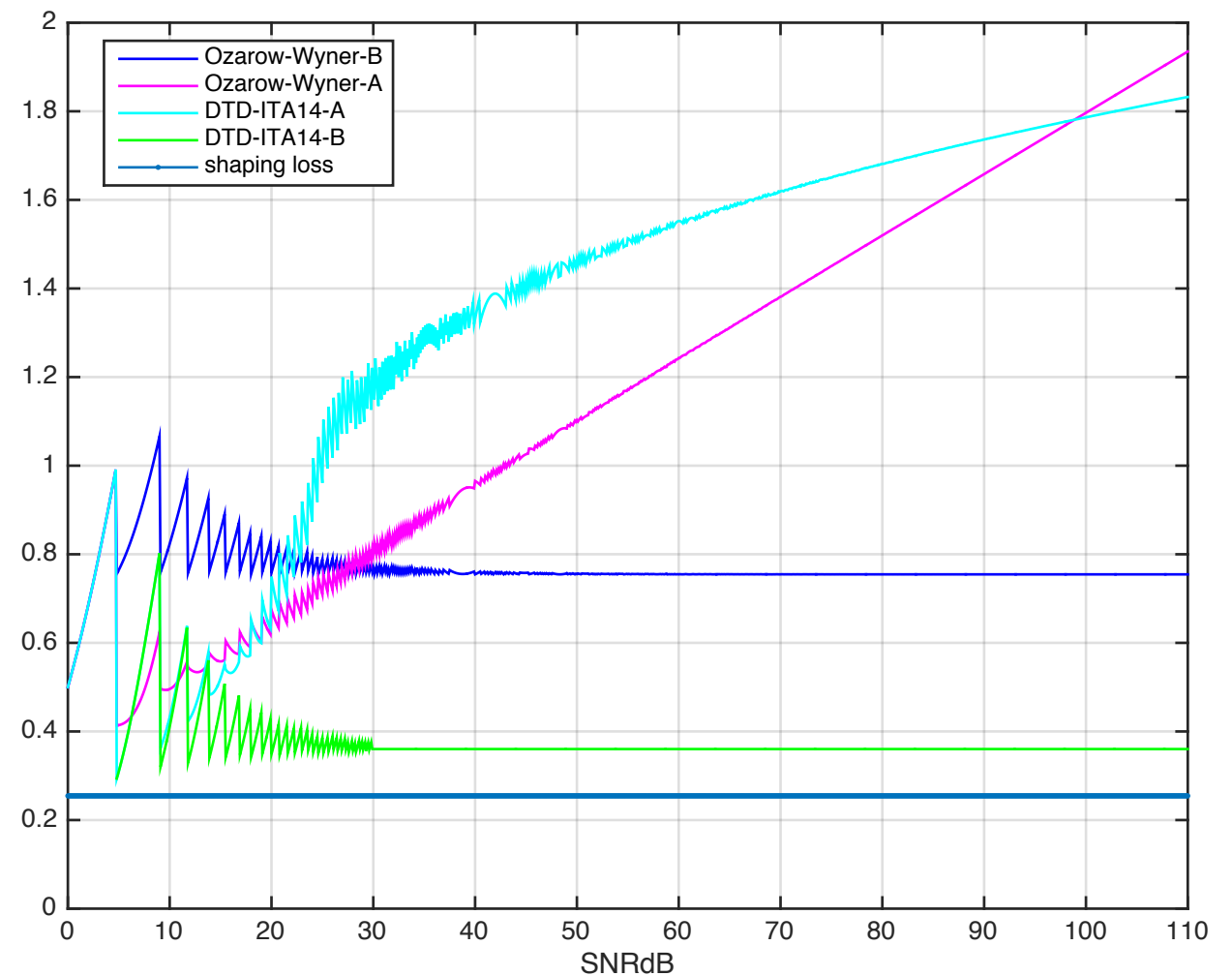
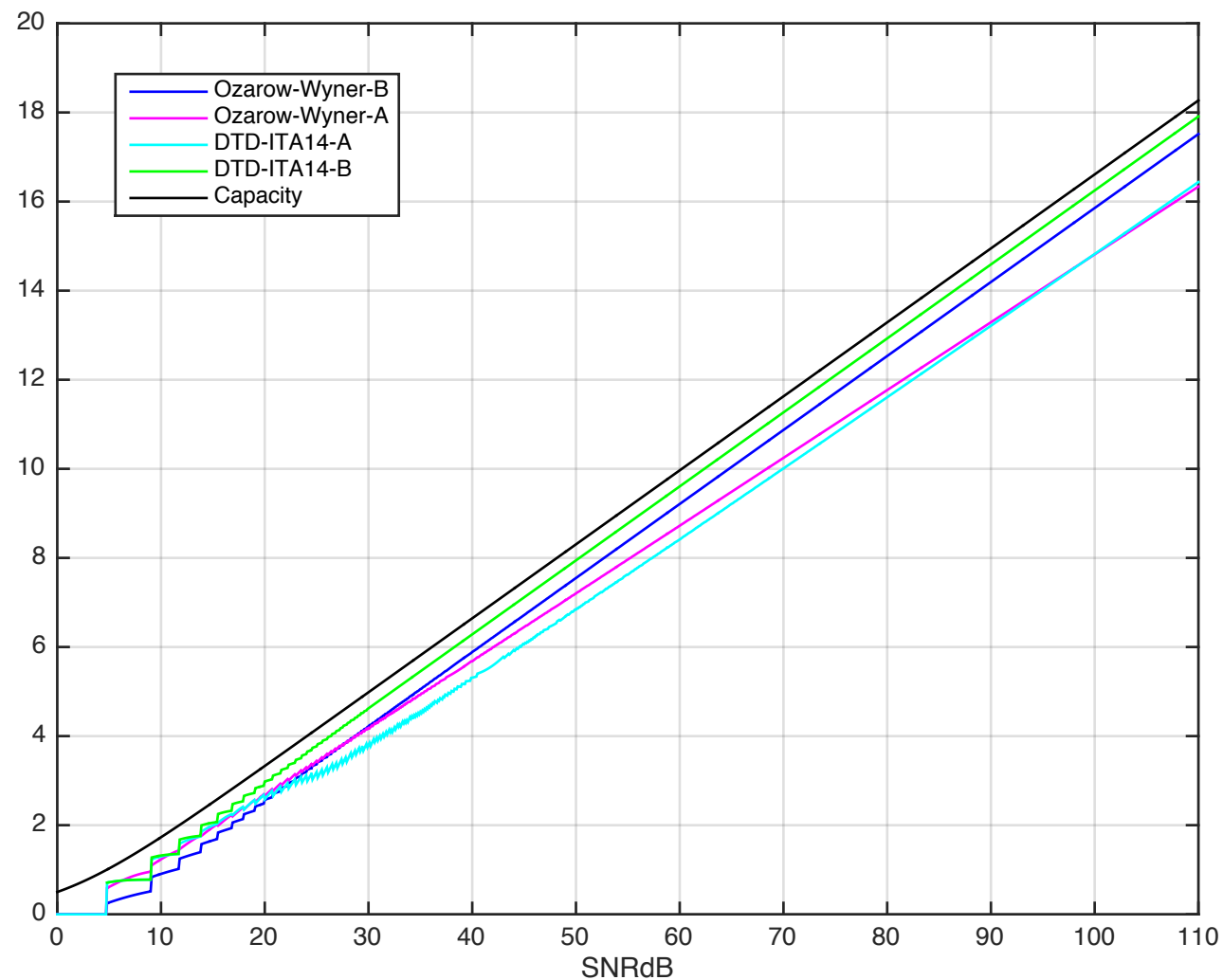
DTD-ITA`14-B

$$I(X_D; \sqrt{S}X_D + Z) \geq \log(N) - \text{gap},$$

$$\text{gap} = \frac{1}{2} \log \left(\frac{e}{2} \right) + \frac{1}{2} \log \left(1 + (N - 1) e^{-\frac{Sd_{\min}^2(X_D)}{4}} \right)$$

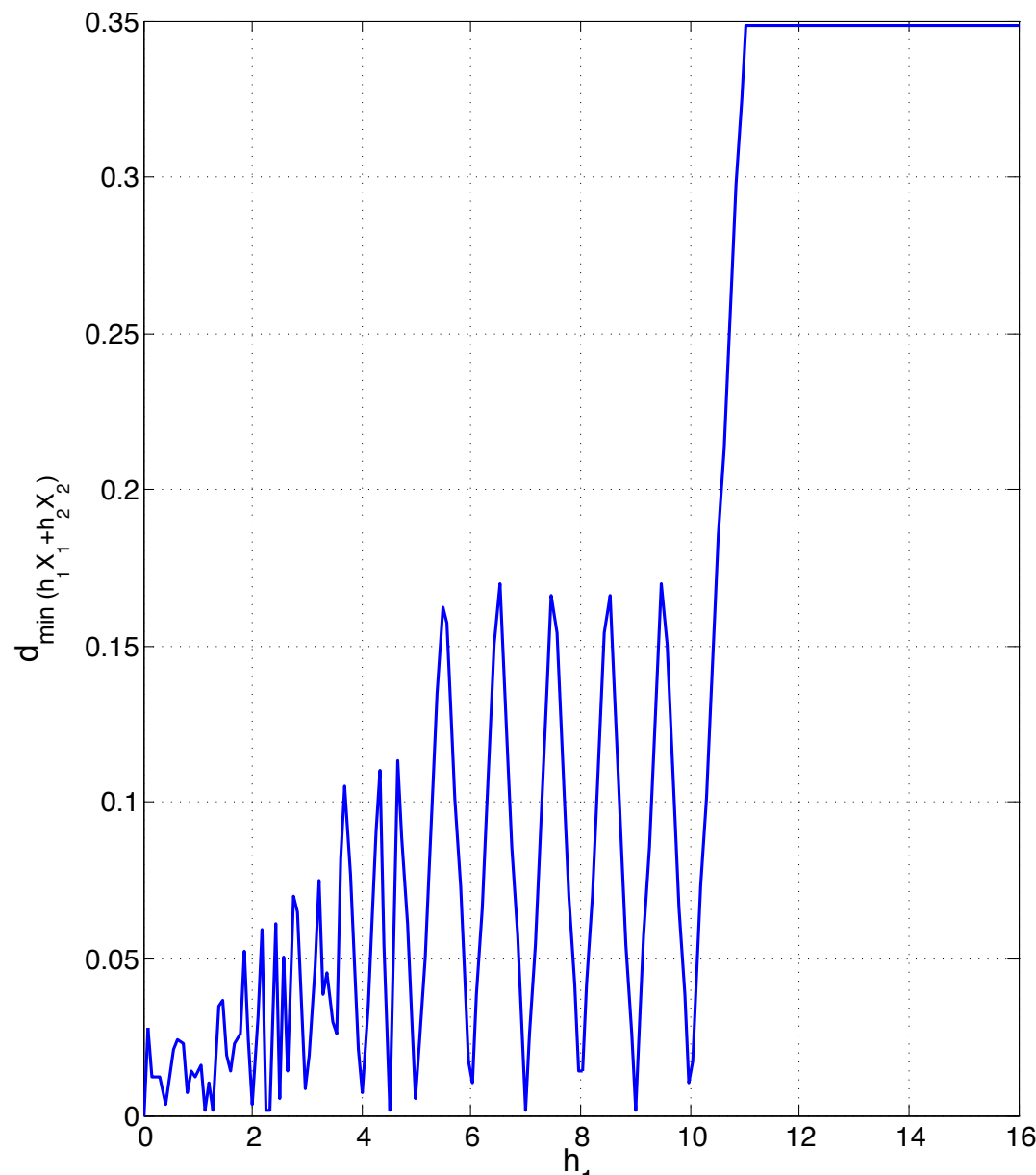
Bound Comparison

Number of Points $N = \lfloor \sqrt{1 + S} \rfloor$



Minimum Distance

Example: $h_2=1$, $N_1=N_2=10$



$$I(X_D; \sqrt{S}X_D + Z) \geq H(X_D) - \text{gap},$$
$$\text{gap} = \frac{1}{2} \log\left(\frac{\pi e}{6}\right) + \frac{1}{2} \log\left(1 + \frac{12}{S d_{\min}^2(X_D)}\right).$$



$$I(X_{1D}, X_{2D}; h_1 X_{1D} + h_2 X_{2D} + Z)$$

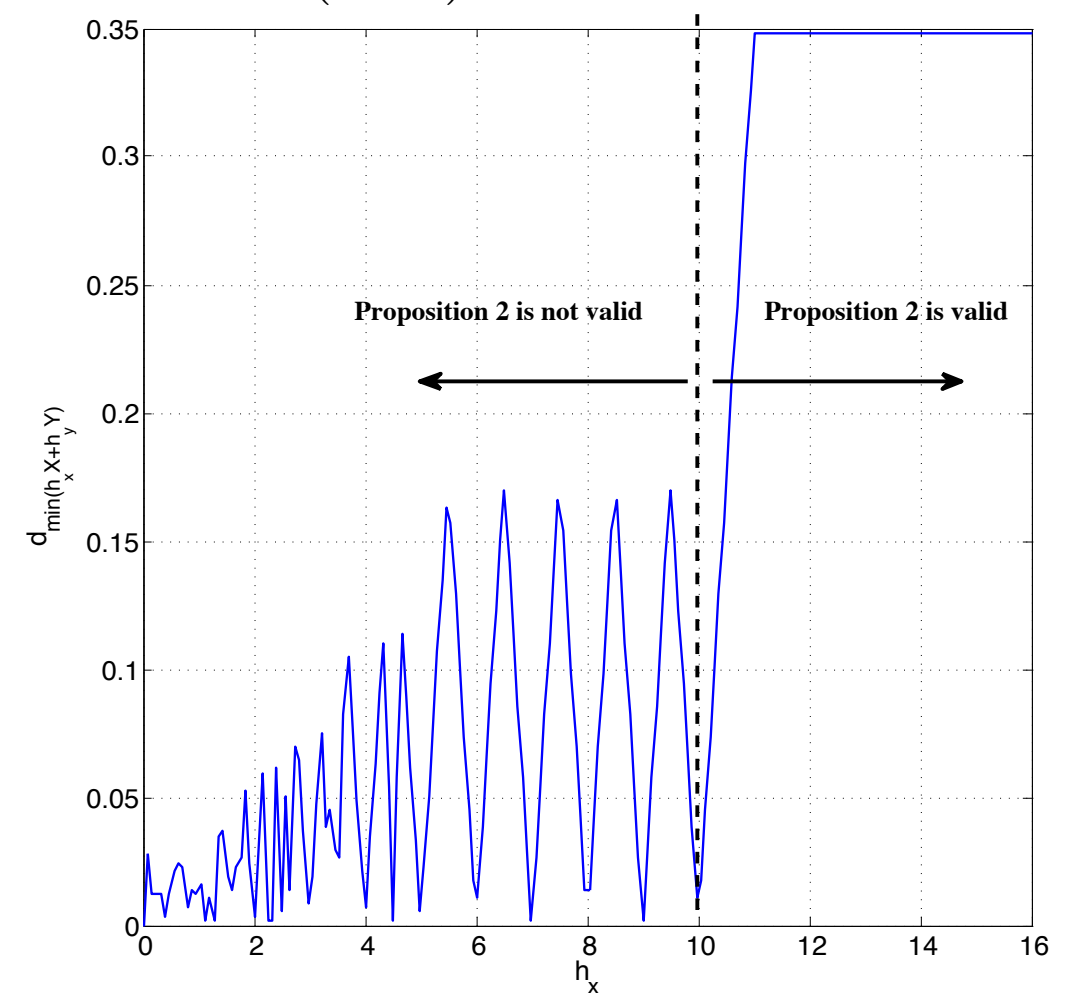
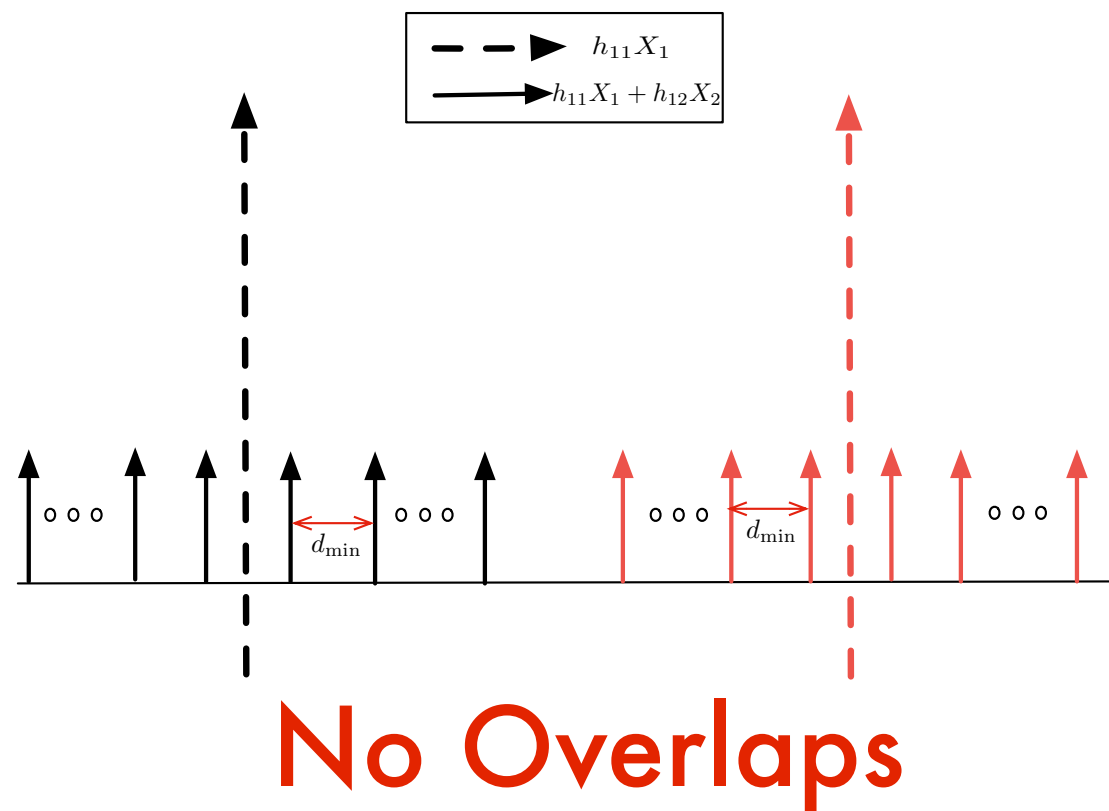
$$d_{\min}(h_1 X_{1D} + h_2 X_{2D})$$

Very Irregular

dmin: bound 1

$$d_{\min}(h_1 X_{1D} + h_2 X_{2D}) = \min(h_1 d_{\min}(X_{1D}), h_2 d_{\min}(X_{2D}))$$

if either $h_2 d_{\min}(X_{2D}) N_2 \leq h_1 d_{\min}(X_{1D})$
 or $h_1 d_{\min}(X_{1D}) N_1 \leq h_2 d_{\min}(X_{2D})$

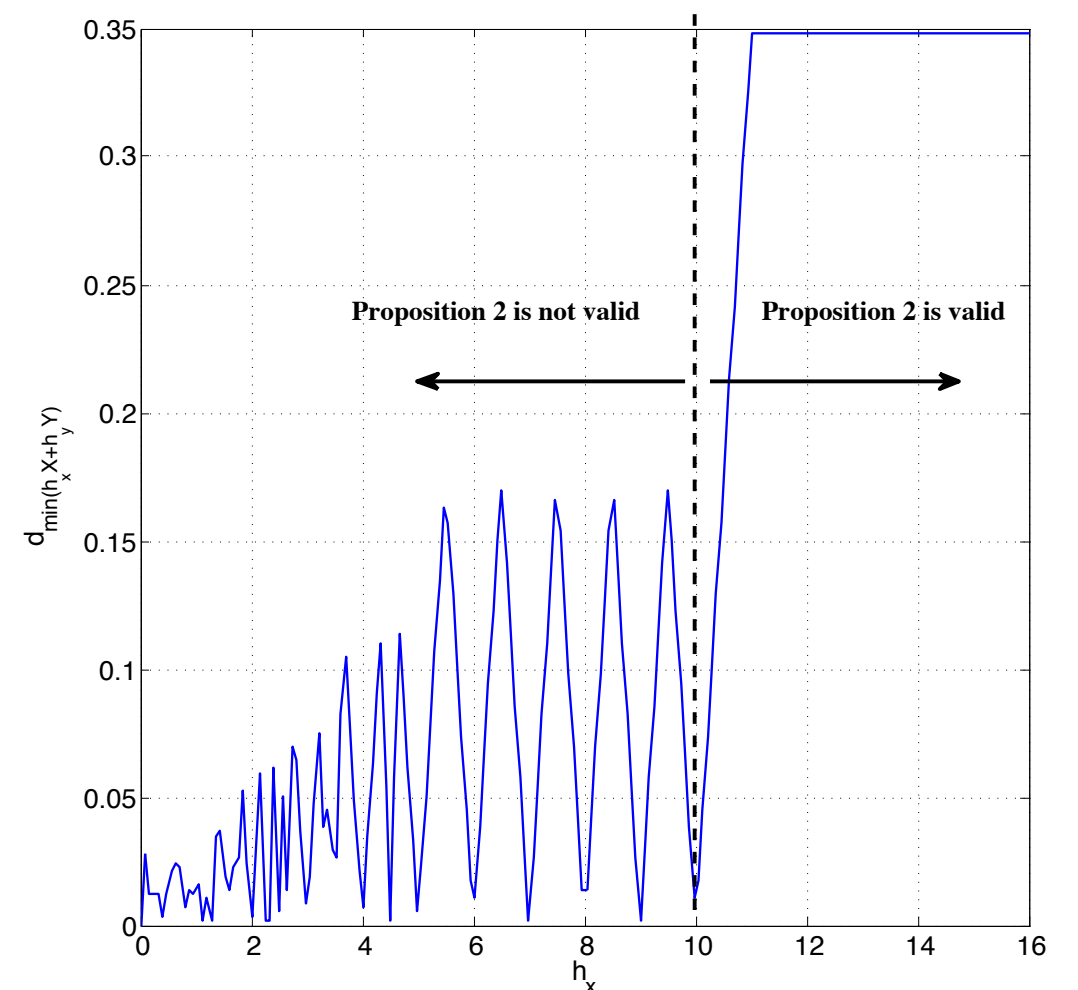
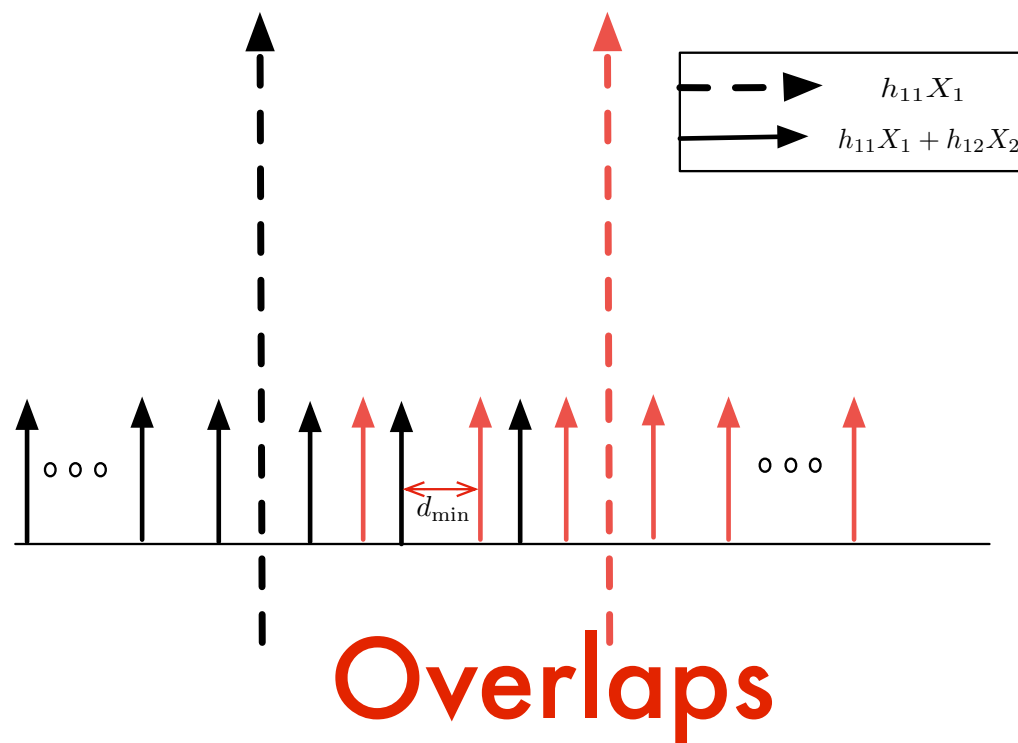


dmin: bound 2

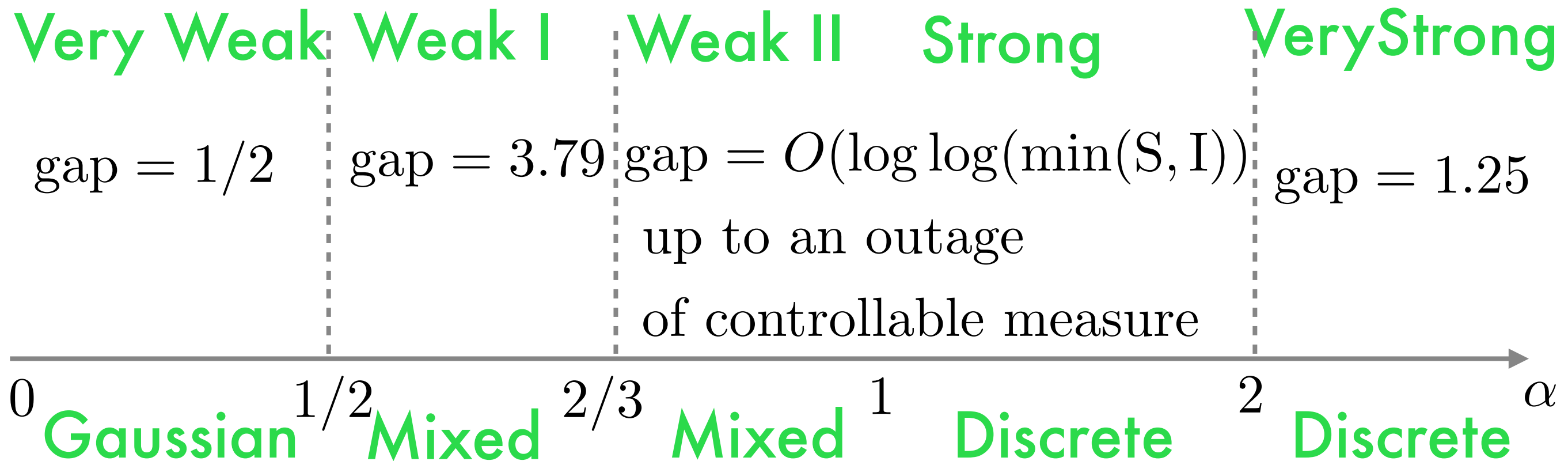
$$d_{\min}(h_1 X_{1D} + h_2 X_{2D}) = \kappa \min \left(h_1 d_{\min}(X_{1D}), h_2 d_{\min}(X_{2D}), \max \left(\frac{h_1 d_{\min}(X_{1D})}{N_2}, \frac{h_2 d_{\min}(X_{2D})}{N_1} \right) \right)$$

$$\kappa = \frac{\gamma}{(1 + \log(\max(N_1, N_2)))}$$

for all (h_1, h_2) except an outage set of measure γ for any $\gamma > 0$.



Main Result

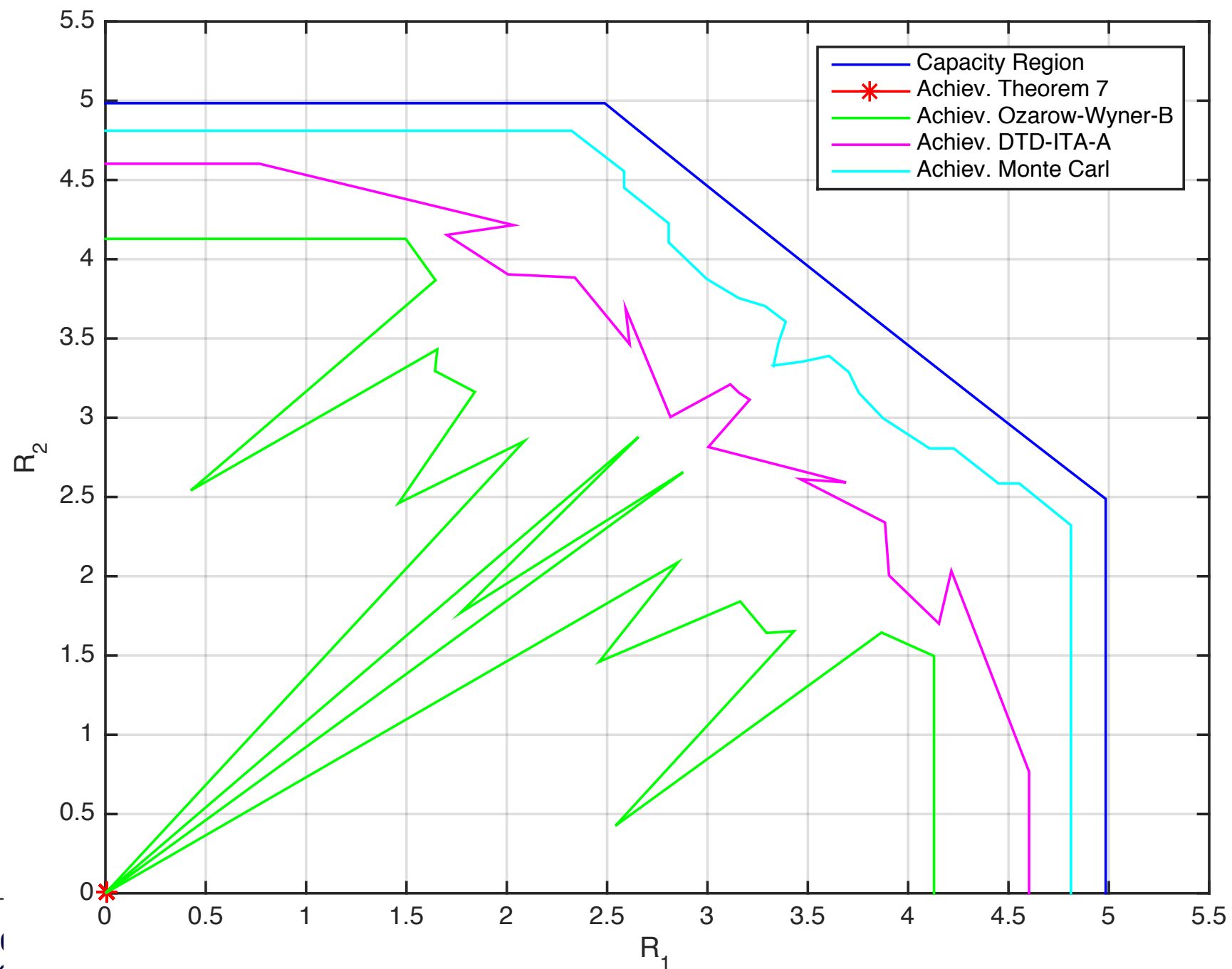


$$X_i = \sqrt{1 - \delta_i} X_{iD} + \sqrt{\delta_i} X_{iG}, \quad i \in [1 : 2],$$

$$\alpha = \frac{\log I}{\log S}$$

Closed-form expressions
for number of points,
power splits and gap

Example



PAM+PAM

$$X_D := X_{D_1} + X_{D_2},$$

where $X_{D_1} \sim \text{discrete} : d_{\min}(X_{D_1}) > 0$,

$X_{D_2} \sim \text{discrete} : d_{\min}(X_{D_2}) > 0$,

$$X_M := X_{D_1} + X_G,$$

where X_{D_1}, X_G and X_{D_2} are mutually independent. Then, for $Z_G \sim \mathcal{N}(0, 1)$ independent of everything else, we have

$$I(X_D; gX_D + Z_G) - I(X_M; gX_M + Z_G) \leq \frac{1}{2} \log(2),$$

$$I(X_M; gX_M + Z_G) - I(X_D; gX_D + Z_G) \leq \frac{1}{2} \log\left(\frac{\pi e}{3}\right) + \frac{1}{2} \log\left(1 + \frac{12}{g^2 d_{\min}^2(X_D)}\right).$$

Concluding Remarks

- Key idea: use non-Gaussian inputs
- Developed very general tools of use beyond 2-IC
- Applicable to Block Asynchronous Interference Channel and Codebook Oblivious Interference Channel

Thank you

arXiv:1506.02597

odytso2@uic.edu